
AIM-FM: Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond

Abstract

Towards next-generation medical analysis: Unlock the potential of medical foundation models for more explainable, robust, secure diagnosis solutions

Workshop Homepage: <https://wymancv.github.io/AIM-FM-NeurIPS-Workshop/>

Main Proposal

Introduction There have been notable advancements in large foundation models (FMs) [9, 8, 10, 14, 15, 12], which exhibit generalizable language understanding, visual recognition, and audio comprehension capabilities. These advancements highlight the potential of personalized AI assistants in efficiently assisting with daily tasks, ultimately enhancing human life.

Healthcare is one of the most crucial industries touching every individual. Yet, due to large populations and limited medical professionals, it faces significant challenges, including the high cost and low doctor-to-population ratio [1, 3, 2, 5]. This shortage is more pronounced in rural and developing regions, where access to qualified doctors is severely limited, exacerbating health disparities and preventing timely treatment for common and complex conditions alike. *Hence, there is a critical need to develop effective, affordable, and professional AI-driven medical assistants.*

Despite the great success in general domains [6, 4, 16], FMs struggle in specific domains requiring strict professional qualifications, such as healthcare [13, 11, 7], which has high sensitivity and security risk. In light of the growing healthcare demands, this workshop aims to explore the potential of Medical Foundation Models (MFMs) in smart medical assistance, thereby improving patient outcomes and streamlining clinical workflows. Considering the primary clinical needs, we emphasize the explainability, robustness, and security of the large-scale multimodal medical assistant, pushing forward its reliability and trustworthiness. By bringing together expertise in diverse fields, we hope to bridge the gap between industry and academia regarding precision medicine, highlighting clinical requirements, inherent concerns, and AI solutions. Through this cooperative endeavor, we hope to unlock the potential of MFMs, striving for groundbreaking advancements in healthcare.

Objectives With the support of the NeurIPS community, this workshop aims to unleash the potential of cutting-edge MFMs in enhancing medical procedures, which is detailed in the following aspects:

- To promote the exploration of multimodal models and data for boosting automatic medical applications and clinical decisions.
- To provide a platform for interdisciplinary collaboration among computer scientists, clinical physicians, and industry professionals, bridging the gap between industry and academia.
- To explore real-world clinical demands, ethical concerns, and advanced technology progress, lightening future research directions in the field of intelligent medicine.
- To showcase the potential of cutting-edge research, development, and adaptation of MFMs to a wide range of medical applications, promoting the fusion of technology and applications.
- To deepen the understanding of how large models can facilitate intuitive control and collaboration for clinical purposes.

- To encourage broad discussions on the current challenges and potential solutions in integrating AI in the healthcare sector.
- To facilitate the fairness and security of the research on medical AI and intelligent healthcare.
- To improve the efficiency and effectiveness of workflows on medical applications and clinical scenarios.

Encouraged Topics This workshop offers an interdisciplinary platform for exceptional researchers to present their latest progress in MFMs regarding healthcare support. Attendees will discuss technological advancements and delve into MFMs-driven diagnostic systems in the healthcare sector. Key topics of interest for the workshop may cover, but are not limited to, the following:

- **MFMs at Scale.** Developing large-scale medical foundation models applicable for hospital use, including diagnosis, prognosis, treatment, surgical assistance, etc.
- **Explainable MFMs.** Attempting to open the black box of MFMs in medical decision-making, ensuring transparency and interpretability in diagnostic outcomes.
- **Robust Diagnosis.** Exploring and enhancing the robustness of MFMs in diverse and challenging medical scenarios, including scarcity/misalignment of medical data, parameter-efficient tuning, validation techniques, etc.
- **Patient Privacy.** Strategies for data/model privacy in tuning and testing MFMs, including federated learning, data encryption, machine unlearning, uncustomization, etc.
- **MFMs with Resource Constraint.** Research on optimizing MFMs with constrained resources, e.g., constrained computation, limited data, limited annotations, etc.
- **Human-AI Interaction.** Studying the interaction dynamics to enhance the collaboration between healthcare professionals/patients and AI, e.g., prompting engineering, feedback refining, system designing, etc.
- **Multimodal Learning.** Research on effectively using heterogeneous medical data for training and addressing multi-modal challenges, such as modality misalignment and missing.
- **Generative Model for Healthcare.** Aiming to develop generative models for producing multimodal data for healthcare, including the generation of medical images, videos, reports, biology structures, etc.
- **Agent for Healthcare.** Towards the applications of AI agent systems in healthcare, including diagnosis, prognosis, surgical assistance, telehealth, etc.
- **Fairness in MFMs.** Targeting to develop fair multimodal models in healthcare and addressing potential bias from data, model, annotation, evaluation, etc.

Attendance Considering that NeurIPS has seen a significant increase in general attendance, the expected number of attendees and other related events are detailed as follows. (1) 150-200 in-person attendees. (2) 200-500 online attendees. (3) 80-100 paper submissions. (4) 20-25 posters.

Tentative List of Invited Speakers The workshop has invited the following in-person keynote speakers with profound experience in medical diagnosis and artificial intelligence.

- **Prof. Animashree Anandkumar** (Email: anima@caltech.edu) (Confirmed)
Bren Professor at California Institute of Technology;
Senior Director of AI Research at NVIDIA
- **Prof. Tianming Liu** (Email: tliu@cs.uga.edu) (Confirmed)
Distinguished Research Professor and Full Professor at University of Georgia
- **Prof. Keane Pearse** (Email: p.keane@ucl.ac.uk) (Confirmed)
Professor at University College London
- **Prof. James Zou** (Email: jamesz@stanford.edu) (Confirmed)
Associate Professor at Stanford University
- **Prof. Faisal Mahmood** (Email: faisalmahmood@bwh.harvard.edu) (Confirmed)
Associate Professor at Harvard University

- **Prof. Sheng Wang** (Email: swang@cs.washington.edu) (Confirmed)
Assistant Professor at University of Washington
- **Dr. Shekoofeh Azizi** (Email: shekazizi@google.com) (Confirmed)
Staff Research Scientist at Google DeepMind

Diversity Statement We have made a great effort to ensure the diversity of keynote speakers and organizers, encompassing various aspects of race, gender, expertise, institution, research areas, etc.

Keynote Speakers. We have curated a diverse lineup of keynote speakers to ensure a broad representation of perspectives and experiences. **(a) Racial Diversity.** Our speakers come from various racial and ethnic backgrounds, including Asian, American, European, and Indian, reflecting the global nature of the medical research community. **(b) Gender Diversity.** Our speaker includes male and female researchers, ensuring the talk is free of gender bias. **(c) Expertise Diversity.** Our speakers represent a wide range of research areas within the medical domain, including clinical medicine, computational pathology, computational biology, medical image analysis, biomarker discovery, etc. This diversity of expertise ensures that the workshop covers a comprehensive range of topics and applications related to medical foundation models. **(d) Institution Diversity.** In addition to researchers from academic institutions, we have invited speakers from leading industry companies, such as Google and NVIDIA, to provide insights into the practical applications of medical foundation models in real-world settings.

Organizers. We have assembled a diverse team to ensure a broad range of perspectives and experiences. **(a) Racial Diversity.** Our organizers come from different racial and ethnic backgrounds, including Asian, European, and North American. This diversity in race and ethnicity brings a richness of cultural viewpoints and experiences to the workshop organization. **(b) Gender Diversity.** Our organizers comprise a balance of male and female researchers. We believe this gender parity fosters a more inclusive viewpoint to addressing the bias issue in medical foundation models. **(c) Expertise Diversity.** Our organizers cover a range of research fields, such as clinical diagnostics, pathology analysis, medical imaging analysis, genomics, drug discovery, and robotic surgery. This diverse expertise guarantees that the workshop addresses various medical needs and concerns. **(d) Area Diversity.** Our team comprises biologists with expertise in biomedicine and computer scientists focused on developing general artificial intelligence. In addition, we have research specialists dedicated to advancing AI-assisted medical applications and clinical practice.

Earlier Versions As one of the emerging fields, no prior workshops have specifically addressed the topic of medical foundation models, emphasizing the novelty and growing interest in this field.

Points of Difference The AIM-FM workshop, focusing on Advancements in Medical Foundation Models, is distinguished by three key aspects. **(a) Pioneering Focus.** AIM-FM is the first workshop dedicated to large-scale medical foundation models. This unique focus highlights its pioneering status in addressing critical and complex clinical challenges for the next-generation medical diagnosis, which has unique clinical values different from previous workshops. It also represents the first attempt to delve deeply into three fundamental yet often underexplored clinical themes, including explainability, robustness, and security, which are essential for the next-generation medical models. **(b) Interdisciplinary Collaboration.** By assembling experts from various fields, including healthcare professionals, AI researchers, biologists, and industry leaders, this workshop first fosters a multidisciplinary dialogue essential for comprehensive advancements in precision medicine. It also provides a platform for exchanging cutting-edge research and real-world applications, ensuring that academic discoveries are translated into practical solutions. **(c) Impact on Healthcare Innovation.** The workshop is poised to drive significant advancements in medical AI, which has great potential to set new standards and benchmarks for future research and revolutionize healthcare methodology.

Technical Needs No other special technical needs.

Other Comments This workshop aims to delve into the profound impact of large-scale medical foundation models on healthcare, particularly in the context of medical assistance. Our primary goal is to foster the development of intelligent, adaptive, and collaborative medical assistants that can learn from vast amounts of multimodal data. By bringing together diverse researchers worldwide, we hope to enhance the precision, safety, and efficiency of MFMs-based medical diagnosis. We emphasize the critical aspects of explainability, robustness, and security in medical AI assistance, as they are crucial for gaining trust and ensuring the ethical use of AI in healthcare.

Organization Information

Organizer List The organizer list is as follows.

1. Prof. Yixuan Yuan (Email: yxyuan@ee.cuhk.edu.hk)

Research Area. Prof. Yuan is an Assistant Professor at the Department of Electronic Engineering at the Chinese University of Hong Kong. Her research spans several key areas in medical AI technology, including medical image analysis, deep learning applications in healthcare, abnormality detection, and developing multimodal medical foundation models. Her work focuses on diverse medical contexts such as brain imaging, endoscopy, dermatoscopy, ophthalmic imaging, and dental diagnostics. Prof. Yuan has authored more than 200 papers, contributing significantly to the field of precision medicine. She received several premium awards, including the CVPR 2022 Best Paper Finalist, MICCAI 2022 Young Scientist Award, 2018 Young Scientist Award from the Hong Kong Institute of Science, etc.

Organization Experience. Prof. Yuan has successfully organized several significant workshops, including EARTH: Embodied AI and Robotics for Healthcare workshop at MICCAI 2024, Vision Meets Drones challenge at ICCV 2021, Deep Generative Models workshop at MICCAI 2020-2024. She also serves as Conference Organization Committee of ISBI 2025, ICRA 2021, Area Chair of MICCAI 2019-2022. Additionally, Prof. Yuan also participates in organizing several special issues in top journals, including Advancements in Foundation Models for Medical Imaging special issue of IEEE TMI 2024, and Advancements in Foundation Models special issue of IEEE TNNLS 2024. Prof. Yuan's broad experience in managing high-profile academic events and her leadership in AI research ensure she is exceptionally well-prepared to contribute to the success of the NeurIPS workshop.

2. Prof. Qin Yao (Email: yaoqin@ucsb.edu)

Research Area. Prof. Yao is an Assistant Professor of the Department of Electrical and Computer Engineering at the University of California, Santa Barbara, and a Senior Research Scientist at Google DeepMind, where she works on Gemini Safety. Her research is centered on enhancing the robustness of machine learning systems. She is deeply engaged in developing general machine learning algorithms with broad applications across computer vision, natural language processing, and healthcare, particularly managing diabetes. Her work extends into AI safety. She has obtained the Adobe Faculty Research Award, and the NeurIPS Women in Machine Learning Travel Award, etc.

Organization Experience. She successfully organized R0-FoMo: Robustness of Few-shot and Zero-shot Learning in Foundation Models workshop at NeurIPS 2023, Southern California Data Science Day workshop at KDD 2023, and serves as Area Chair of Women in Machine Learning (WiML) Workshop 2019-2022. She has rich experience in organizing high-profile workshops, including past events at NeurIPS, proving the ability to organize the upcoming workshop successfully.

3. Prof. Xiang Li (Email: xiangli.shaun@gmail.com)

Research Area. Prof. Li is an Assistant Professor at the Massachusetts General Hospital and Harvard Medical School, Department of Radiology, and an Affiliate Faculty Member at the Kempner Institute for Natural and Artificial Intelligence of Harvard University. Li's primary research interest is medical data analysis for clinical practice, focusing on clinical workflow streamlining, multimodal multi-scale image fusion, and foundation models for healthcare. His research in medical foundation models and deep learning for healthcare has been awarded by multiple federal and industry partners, including NIH, the NVidia Global Impact Award, and the Google Research Scholar Program.

Organization Experience. He successfully organized the Multiscale Multimodal Medical Imaging (MMMI) Workshop, the SAGES Critical View of Safety Challenge and the Multimodal Learning and Fusion Across Scales for Clinical Decision Support workshop at MICCAI. Prof. Li also serves as an area chair of MICCAI 2023-2024, The SAGES Critical View of Safety Challenge 2024, NeurIPS Workshop on Machine Learning in Computational Biology 2019, ACM SIGKDD Workshop on Mining and Learning from Time Series 2019-2022.

4. Prof. Ying Wei (Email: ying.wei@ntu.edu.sg)

Research Area. Prof. Wei is an Assistant Professor at the School of Computer Science and Engineering at Nanyang Technological University. She works on machine learning and is especially interested in solving problems with few-shot data via knowledge transfer. These problems include drug discovery, drug synthesis, and medical image analysis. She has published over 55 papers in conferences such as ICML and NeurIPS. She was supported by the Hong Kong PhD Fellowship. She has received the ICLR 2024 Honorable Mention and SIGKDD 2014 Best Paper nomination.

Organization Experience. She has served as an action editor for TMLR, an area chair for ICML and NeurIPS, a senior program committee member for AAAI, and a committee member for conferences

such as NeurIPS and ICLR. She has also organized the First Workshop on Pre-training: Perspectives, Pitfalls, and Paths Forward at ICML 2022, which has shared expertise with the NeurIPS community. Her pivotal role highlights her commitment to addressing key challenges in large model pre-training.

5. Prof. Bulat Ibragimov (Email: bulat@di.ku.dk)

Research Area. Prof. Ibragimov is an Associate Professor of the Department of Computer Science at the University of Copenhagen. His research interests include machine learning in medicine, computer-aided diagnosis, medical image analysis, and human-AI interaction. He received various awards, including the Novo Nordisk Award for Young Data Science Investigator.

Related Events. He has won three international competitions for developing AI-based solutions for computerized cephalometry, vertebra annotation, and intervertebral disc annotation. He has 7.5 years of research experience in the field of medical imaging and machine learning for musculoskeletal, cardiac, dental, and radiotherapy planning applications.

6. Prof. Linda Petzold (Email: petzold@engineering.ucsb.edu)

Research Area. Prof. Petzold is a Mehrabian Distinguished Professor of Mechanical Engineering and Computer Science at the University of California, Santa Barbara. Her research focuses on modeling, analysis, simulation, and software applied to multiscale, networked systems in biology, materials, and social networks. Her research group has been developing advanced algorithms for discrete stochastic simulation of systems where the fate of a few key molecules can make a big difference in important outcomes. She was awarded the Sidney Fernbach Award by the IEEE Computer Society, etc.

Organization Experience. She has achieved significant academic achievement and successfully organized lots of workshops, including the DOE Multiscale Workshop 2004, UCSB Workshop on Control and Computation 1998, ARO Workshop on Real-Time Simulation of Multibody Systems 1992. She is the organizer of SIAM Conf. on Computational Science and Engineering 2003, First Southern California Applied Mathematics Symposium 2021, SIAM Conf. on Computational Science and Engineering 2000, Sci-CADE01, Scientific Computing, and Differential Equations 2001, Oberwolfach Conference on Numerical Methods for Differential-Algebraic Equations 1995.

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